Expected j-th Order Statistic Under Uniform Sampling Without Replacement

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1 Setup and Notation

Let $x_1, \dots, x_N \in \mathbb{R}$ be a fixed batch of values. Let

$$x_{(1)} \le x_{(2)} \le \cdots \le x_{(N)}$$

denote the sorted values (assume they are distinct for simplicity; ties can be handled by grouping equal values and working with ranks-with-multiplicity).

We sample a subset $S \subset \{1, \dots, N\}$ of size k *uniformly without replacement*:

$$|S| = k, \qquad \mathbb{P}(S) = \binom{N}{k}^{-1}.$$

Let $X_{(1:k)} \le \cdots \le X_{(k:k)}$ be the order statistics of the selected values $\{x_i : i \in S\}$. Fix an index $j \in \{1, \dots, k\}$. Our goal is to compute:

1. the unconditional expectation $\mathbb{E}[X_{(j:k)}]$;
2. the conditional expectation $\mathbb{E}[X_{(j:k)} \mid i \in S]$ for a fixed member i;
3. the leave-one-out expectation $\mathbb{E}[X^{(-i)}_{(j:k)}]$, where the population removes member i and then samples k from the remaining $N-1$ values.

Throughout we use the convention $\binom{a}{b} = 0$ when $b < 0$ or $b > a$.

2 Unconditional expected j-th order statistic

2.1 Exact probability mass function

For $m \in \{1, \dots, N\}$,

$$\mathbb{P}\big(X_{(j:k)} = x_{(m)}\big) = \frac{\binom{m-1}{j-1}\binom{N-m}{k-j}}{\binom{N}{k}}.$$

Reason. For the j-th smallest selected value to equal $x_{(m)}$:

- $x_{(m)}$ must be selected;
- exactly $j-1$ of the selected values must come from the $m-1$ smaller values $\{x_{(1)}, \dots, x_{(m-1)}\}$;
- the remaining $k-j$ selected values must come from the $N-m$ larger values $\{x_{(m+1)}, \dots, x_{(N)}\}$.

Counting such subsets gives $\binom{m-1}{j-1}\binom{N-m}{k-j}$ possibilities out of $\binom{N}{k}$ total.

2.2 Expectation

Thus,

$$\boxed{\mathbb{E}[X_{(j:k)}] = \sum_{m=1}^{N} x_{(m)} \, \frac{\binom{m-1}{j-1}\binom{N-m}{k-j}}{\binom{N}{k}}}.$$

3 Conditional expected j-th order statistic given $i \in S$

Fix an index $i \in \{1, \dots, N\}$ and let r be its rank so that $x_i = x_{(r)}$. Conditioning on $\{i \in S\}$ is equivalent to: include $x_{(r)}$ and choose the remaining $k-1$ elements uniformly from the other $N-1$ indices. Hence the conditional sample space has size $\binom{N-1}{k-1}$.

3.1 Conditional probability mass function

For each $m \in \{1, \dots, N\}$,

$$\mathbb{P}\big(X_{(j:k)} = x_{(m)} \mid i \in S\big) = \begin{cases} \dfrac{\binom{m-1}{j-1}\binom{N-m-1}{k-j-1}}{\binom{N-1}{k-1}}, & m < r, \\ \dfrac{\binom{r-1}{j-1}\binom{N-r}{k-j}}{\binom{N-1}{k-1}}, & m = r, \\ \dfrac{\binom{m-2}{j-2}\binom{N-m}{k-j}}{\binom{N-1}{k-1}}, & m > r. \end{cases}$$

Reason (case by case).

- **Case $m < r$.** The forced-included element $x_{(r)}$ is *larger* than $x_{(m)}$. For $X_{(j:k)} = x_{(m)}$, we must:
    1. include $x_{(m)}$ (one of the k);
    2. choose $j-1$ from the $m-1$ values below $x_{(m)}$;
    3. choose the remaining $k-j-1$ (excluding the forced $x_{(r)}$) from the values above $x_{(m)}$, of which there are $N-m-1$ after removing $x_{(r)}$.

  This yields $\binom{m-1}{j-1}\binom{N-m-1}{k-j-1}$.

- **Case $m = r$.** The forced element is itself the j-th order statistic. Then we need:
    1. choose $j-1$ from the $r-1$ values below it;
    2. choose $k-j$ from the $N-r$ values above it.

  This yields $\binom{r-1}{j-1}\binom{N-r}{k-j}$.

- **Case $m > r$.** The forced-included element $x_{(r)}$ is *smaller* than $x_{(m)}$, so it must be counted among the $j-1$ values below $x_{(m)}$. For $X_{(j:k)} = x_{(m)}$, we must:
    1. include $x_{(m)}$;
    2. choose the remaining $j-2$ values below $x_{(m)}$ from the $m-2$ values in $\{x_{(1)}, \dots, x_{(m-1)}\} \setminus \{x_{(r)}\}$;

3. choose $k-j$ values from the $N-m$ values above $x_{(m)}$.

This yields $\binom{m-2}{j-2}\binom{N-m}{k-j}$.

3.2 Expectation

Therefore,

$$\boxed{\mathbb{E}[X_{(j:k)} \mid i \in S] = \sum_{m=1}^{N} x_{(m)}\, \mathbb{P}\big(X_{(j:k)} = x_{(m)} \mid i \in S\big)\,,}$$

with the conditional mass given above.

4 Leave-one-out expected j-th order statistic

Now remove member i (rank r), forming a population of size $N-1$:

$$\{x_{(1)}, \ldots, x_{(r-1)}, x_{(r+1)}, \ldots, x_{(N)}\}.$$

Assume $k \leq N-1$ so sampling k items is feasible.

Let $X_{(j:k)}^{(-i)}$ denote the j-th order statistic of a uniform k-subset drawn from these $N-1$ values.

4.1 Weights on the reduced population

For a reduced population of size $N-1$, the unconditional weight for the p-th smallest reduced value is

$$w_p^- = \mathbb{P}\Big(X_{(j:k)}^{(-i)} = x_{(p)}^{(-i)}\Big) = \frac{\binom{p-1}{j-1}\binom{(N-1)-p}{k-j}}{\binom{N-1}{k}}, \qquad p = 1, \ldots, N-1.$$

4.2 Expressing in terms of the original sorted list

Removing rank r shifts ranks above r down by one:

$$x_{(p)}^{(-i)} = \begin{cases} x_{(p)}, & p < r, \\ x_{(p+1)}, & p \geq r. \end{cases}$$

Hence

$$\boxed{\mathbb{E}\Big[X_{(j:k)}^{(-i)}\Big] = \sum_{m=1}^{r-1} x_{(m)}\, w_m^- + \sum_{m=r+1}^{N} x_{(m)}\, w_{m-1}^-,}$$

where $w_p^- = \dfrac{\binom{p-1}{j-1}\binom{(N-1)-p}{k-j}}{\binom{N-1}{k}}$.

5 Matrix recipes for computing all $j = 1, \ldots, k$ simultaneously

This section rewrites the previous scalar formulas in a way that computes *all* expectations $\{\mathbb{E}[X_{(j:k)}]\}_{j=1}^{k}$ simultaneously using matrix operations. The key point is that for fixed (N, k) the relevant *weight matrices* depend only on combinatorial coefficients and can be precomputed once and reused.

5.1 Unconditional expectations for all j

Define the weight matrix $W \in \mathbb{R}^{N\times k}$ by

$$W_{m,j} = \frac{\binom{m-1}{j-1}\binom{N-m}{k-j}}{\binom{N}{k}}, \qquad m = 1,\dots,N, \;\; j = 1,\dots,k. \tag{1}$$

Let $x \in \mathbb{R}^N$ be the vector of sorted values:

$$x = (x_{(1)}, x_{(2)}, \dots, x_{(N)})^\top.$$

Let $E \in \mathbb{R}^k$ collect the expectations across j:

$$E = \big(\mathbb{E}[X_{(1:k)}], \dots, \mathbb{E}[X_{(k:k)}]\big).$$

Then

$$\boxed{E = x^\top W.} \tag{2}$$

Thus computing all k order-statistic expectations reduces to a single matrix–vector multiply (after sorting x).

5.2 Conditional expectations for all j and all included ranks

Fix a rank $r \in \{1,\dots,N\}$ and condition on including the element of rank r (i.e. include $x_{(r)}$). Define three $N \times k$ matrices A, B, C (all depending only on N, k) as follows:

$$A_{m,j} = \frac{\binom{m-1}{j-1}\binom{N-m-1}{k-j-1}}{\binom{N-1}{k-1}}, \tag{3}$$

$$B_{m,j} = \frac{\binom{m-1}{j-1}\binom{N-m}{k-j}}{\binom{N-1}{k-1}}, \tag{4}$$

$$C_{m,j} = \frac{\binom{m-2}{j-2}\binom{N-m}{k-j}}{\binom{N-1}{k-1}}. \tag{5}$$

(As usual, invalid binomials evaluate to zero.)

For a fixed included rank r, the conditional pmf from the previous section can be viewed as a *row-wise splice*:

$$\mathbb{P}(X_{(j:k)} = x_{(m)} \mid \text{include rank } r) = \begin{cases} A_{m,j}, & m < r, \\ B_{r,j}, & m = r, \\ C_{m,j}, & m > r. \end{cases}$$

Hence the conditional expectation for a fixed r is

$$\mathbb{E}[X_{(j:k)} \mid \text{include rank } r] = \sum_{m<r} x_{(m)} A_{m,j} + x_{(r)} B_{r,j} + \sum_{m>r} x_{(m)} C_{m,j}.$$

Vectorized computation for all r at once. Let $\mathbf{1}_k \in \mathbb{R}^k$ be the all-ones vector and form the replicated data matrix $X \in \mathbb{R}^{N\times k}$:

$$X := x\,\mathbf{1}_k^\top, \qquad \text{so that } X_{m,j} = x_{(m)}.$$

Let $\odot$ denote the Hadamard (elementwise) product. Define

$$P_A := X \odot A, \qquad P_C := X \odot C.$$

Define a row-wise cumulative-sum operator $\mathrm{cumsum}(\cdot)$ that maps an $N \times k$ matrix to another $N \times k$ matrix by

$$\big(\mathrm{cumsum}(M)\big)_{r,j} = \sum_{m=1}^{r} M_{m,j}.$$

Also define the *shifted* cumulative sums with a leading zero row:

$$\widehat{\mathrm{cumsum}}(M) := \begin{bmatrix} 0 \\ \mathrm{cumsum}(M) \end{bmatrix} \in \mathbb{R}^{(N+1)\times k}, \quad \text{so that} \quad \widehat{\mathrm{cumsum}}(M)_{r,j} = \sum_{m=1}^{r-1} M_{m,j}.$$

Now define the matrix $E^{\mathrm{inc}} \in \mathbb{R}^{N\times k}$ whose r-th row collects all conditional expectations given inclusion of rank r:

$$E^{\mathrm{inc}}_{r,j} := \mathbb{E}[X_{(j:k)} \mid \text{include rank } r].$$

Then the entire matrix E^{inc} can be computed by:

$$\boxed{E^{\mathrm{inc}} = \underbrace{\widehat{\mathrm{cumsum}}(P_A)_{1:N,:}}_{\sum_{m<r} x_{(m)} A_{m,:}} + \underbrace{X \odot B}_{x_{(r)} B_{r,:}} + \underbrace{\Big(\mathrm{cumsum}(P_C)_{N,:}\mathbf{1}_N^\top\Big)^\top - \mathrm{cumsum}(P_C)}_{\sum_{m>r} x_{(m)} C_{m,:}}} \tag{6}$$

where $\mathbf{1}_N$ is the all-ones vector of length N, and $\mathrm{cumsum}(P_C)_{N,:}$ denotes the last row (a $1 \times k$ row vector) replicated down N times.

In words:

- prefix-sum P_A to get all $\sum_{m<r} x_{(m)} A_{m,:}$,
- add the diagonal contribution $x_{(r)} B_{r,:}$ rowwise,
- add a suffix-sum of P_C to get all $\sum_{m>r} x_{(m)} C_{m,:}$.

This is $O(Nk)$ time after precomputing A, B, C.

5.3 Leave-one-out expectations for all j and all removed ranks

Now remove rank r (i.e. remove $x_{(r)}$) and draw k from the remaining $N-1$ values. Precompute the reduced-population weight matrix $W^- \in \mathbb{R}^{(N-1)\times k}$:

$$W^-_{p,j} = \frac{\binom{p-1}{j-1}\binom{(N-1)-p}{k-j}}{\binom{N-1}{k}}, \qquad p = 1, \ldots, N-1, \;\; j = 1, \ldots, k. \tag{7}$$

Let $E^{\mathrm{loo}} \in \mathbb{R}^{N\times k}$ collect the leave-one-out expectations by removed rank:

$$E^{\mathrm{loo}}_{r,j} := \mathbb{E}[X^{(-r)}_{(j:k)}],$$

where $X^{(-r)}_{(j:k)}$ is the j-th order statistic of a k-subset drawn from $\{x_{(1)}, \ldots, x_{(r-1)}, x_{(r+1)}, \ldots, x_{(N)}\}$.

The rank-shift identity implies:

$$E^{\text{loo}}_{r,:} = \sum_{m<r} x_{(m)} W^{-}_{m,:} + \sum_{m>r} x_{(m)} W^{-}_{m-1,:}.$$

Define two length-$(N-1)$ vectors and their replicated matrices:

$$u := (x_{(1)}, \ldots, x_{(N-1)})^{\top}, \qquad v := (x_{(2)}, \ldots, x_{(N)})^{\top},$$

$$U := u\,\mathbf{1}_k^{\top} \in \mathbb{R}^{(N-1)\times k}, \qquad V := v\,\mathbf{1}_k^{\top} \in \mathbb{R}^{(N-1)\times k}.$$

Define

$$P_1 := U \odot W^{-}, \qquad P_2 := V \odot W^{-}.$$

Let $\widehat{\text{cumsum}}(\cdot)$ be the shifted cumulative sum defined above, now producing an $N \times k$ matrix when applied to an $(N-1) \times k$ matrix.

Then all leave-one-out expectations can be computed simultaneously as

$$\boxed{E^{\text{loo}} = \widehat{\text{cumsum}}(P_1) + \left(\text{cumsum}(P_2)_{N-1,:} \mathbf{1}_N^{\top}\right)^{\top} - \widehat{\text{cumsum}}(P_2).} \tag{8}$$

Here $\text{cumsum}(P_2)_{N-1,:}$ is the last row of $\text{cumsum}(P_2)$, replicated down N times.

Summary of precomputation vs. per-batch work. For fixed (N, k), precompute W, A, B, C, and W^{-} once. For each new batch of values, sort to obtain x, form $X = x\mathbf{1}_k^{\top}$, and compute:

$$E = x^{\top} W, \qquad E^{\text{inc}} \text{ via } (6), \qquad E^{\text{loo}} \text{ via } (8).$$

The conditional and leave-one-out computations are $O(Nk)$ and can be implemented with elementwise multiplies and rowwise prefix sums (cumulative sums), i.e. without looping over j or r.

6 Connections to L-statistics and coverage objectives

6.1 L-statistics viewpoint and why it matters

An *L-statistic* is any linear combination of order statistics computed from a sample. For a size-k subset S, an L-statistic has the form

$$T(S) \;=\; \sum_{j=1}^{k} a_j\, X_{(j:k)}, \tag{9}$$

for some coefficients $a_1, \ldots, a_k$ (possibly depending on k). Common special cases include:

- **Trimmed mean:** average of the middle order statistics after discarding the smallest and largest fractions.
- **Winsorized mean:** clamp extremes and then average, also expressible as a linear combination of order statistics.
- **Median and quantiles:** single order statistics (a degenerate L-statistic).

- **Top-t / bottom-t averages:** averages of the largest or smallest t order statistics.

The key observation for this note is that, under uniform sampling without replacement from a *fixed* finite batch $x_{(1)} \le \cdots \le x_{(N)}$, the expectation of *each* sample order statistic is itself a linear functional of the *population* order statistics:

$$\mathbb{E}[X_{(j:k)}] \;=\; \sum_{m=1}^{N} x_{(m)}\, W_{m,j}, \qquad W_{m,j} = \frac{\binom{m-1}{j-1}\binom{N-m}{k-j}}{\binom{N}{k}}.$$

Consequently, the expectation of any L-statistic (9) is also a linear functional of the sorted batch:

$$\mathbb{E}[T(S)] \;=\; \sum_{j=1}^{k} a_j\, \mathbb{E}[X_{(j:k)}] \;=\; \sum_{m=1}^{N} x_{(m)} \left(\sum_{j=1}^{k} a_j\, W_{m,j} \right). \tag{10}$$

In matrix form, if $W \in \mathbb{R}^{N\times k}$ is the unconditional weight matrix and $a = (a_1, \ldots, a_k)^\top$, then

$$\mathbb{E}[T(S)] \;=\; x^\top (Wa).$$

This has two practical implications:

1. **Precomputation:** for fixed (N, k) and a chosen L-statistic coefficients a, the combined weight vector $w^{(a)} := Wa \in \mathbb{R}^N$ can be precomputed once. Each evaluation reduces to sorting x and one dot product $x^\top w^{(a)}$.

2. **Differentiable surrogates:** many robust objectives used in practice (e.g. trimmed or winsorized means) are L-statistics, so their expectations under random subset selection admit a closed-form linear-in-$x_{(m)}$ representation. This is useful for analysis (bias/variance calculations) and for efficient evaluation inside larger pipelines (e.g. repeated subset sampling).

All of the conditional and leave-one-out constructions in Section 5 inherit the same property: for each fixed conditioning event ("include rank r" or "remove rank r"), the conditional expectation $\mathbb{E}[X_{(j:k)} \mid \cdot]$ remains a linear functional of the sorted batch values, with weights that depend only on (N, k) (and the rank r). Therefore any conditional/leave-one-out L-statistic can be computed by combining those weights exactly as in (10).

6.2 Set coverage as a sum of binary max objectives

Now suppose each sample i has a set of *properties* (or "concepts") from a universe $\mathcal{U} = \{1, \ldots, q\}$. Let

$$b_i^{(u)} \;=\; \mathbf{1}\{\text{sample } i \text{ has property } u\} \in \{0, 1\}.$$

For a selected set S of size k, the (unweighted) coverage objective is

$$C(S) \;=\; \sum_{u=1}^{q} \mathbf{1}\{\exists i \in S : \; b_i^{(u)} = 1\}. \tag{11}$$

Since $b_i^{(u)} \in \{0, 1\}$, the event "property u appears at least once" is equivalent to a binary max (logical OR):

$$\mathbf{1}\{\exists i \in S : \; b_i^{(u)} = 1\} \;=\; \max_{i\in S} b_i^{(u)}.$$

Thus coverage decomposes as a sum of per-property max objectives:

$$C(S) \;=\; \sum_{u=1}^{q} \max_{i\in S} b_i^{(u)}. \tag{12}$$

A weighted version is immediate:

$$C_w(S) \;=\; \sum_{u=1}^{q} w_u \max_{i\in S} b_i^{(u)}.$$

Connection to order statistics. For a fixed property u, the multiset $\{b_i^{(u)} : i \in S\}$ consists of k binary values. The max is the k-th order statistic of these binary values:

$$\max_{i\in S} b_i^{(u)} \;=\; B_{(k:k)}^{(u)},$$

where $B_{(1:k)}^{(u)} \le \cdots \le B_{(k:k)}^{(u)}$ are the order statistics of the selected binary indicators. In this sense, set coverage is a *sum of (max-)order-statistic objectives* applied to binary rewards, one per property.

6.3 Expected coverage under uniform sampling without replacement

Let

$$n_u := \sum_{i=1}^{N} b_i^{(u)}$$

be the number of samples in the batch that contain property u. Under a uniform k-subset S, property u is *not* covered iff all selected items come from the $N - n_u$ items with $b_i^{(u)} = 0$. Therefore

$$\mathbb{P}(u \text{ not covered}) = \frac{\binom{N-n_u}{k}}{\binom{N}{k}}, \qquad \mathbb{P}(u \text{ covered}) = 1 - \frac{\binom{N-n_u}{k}}{\binom{N}{k}}.$$

By linearity of expectation (no independence assumptions are needed),

$$\boxed{\mathbb{E}[C(S)] = \sum_{u=1}^{q} \left(1 - \frac{\binom{N-n_u}{k}}{\binom{N}{k}}\right).} \tag{13}$$

The weighted variant is

$$\mathbb{E}[C_w(S)] = \sum_{u=1}^{q} w_u \left(1 - \frac{\binom{N-n_u}{k}}{\binom{N}{k}}\right).$$

Redundancy ("covered at least t times"). If one wants each property to appear at least t times in S, define

$$C_t(S) = \sum_{u=1}^{q} \mathbf{1}\Big\{ \sum_{i\in S} b_i^{(u)} \ge t \Big\}.$$

For each u, the count $\sum_{i\in S} b_i^{(u)}$ is hypergeometric with parameters (N, n_u, k), so $\mathbb{P}(\text{count} \ge t)$ is a hypergeometric tail and $\mathbb{E}[C_t(S)]$ is again the sum of these tail probabilities over u.

Relevance. Coverage objectives and their variants are ubiquitous in subset selection and summarization pipelines: they encode the desire that the selected set "covers" many concepts at least once (or multiple times). Equation (13) shows that under random subset selection the expected coverage depends *only* on the per-property marginal counts n_u, and can be evaluated in $O(q)$ time once the n_u are known (e.g. from a sparse incidence matrix).

7 Exact expected order statistics for sampling with replacement (known p and q)

This section considers an *exact* (non-estimation) setting. There are m possible arms. Pulling arm $a \in \{1, \ldots, m\}$ produces a deterministic value $q_a \in \mathbb{R}$, and arms are sampled i.i.d. with replacement from a known probability vector $p \in \Delta^{m-1}$:

$$\mathbb{P}(A = a) = p_a, \qquad \sum_{a=1}^{m} p_a = 1.$$

We draw k arms i.i.d. $A_1, \ldots, A_k \sim p$, observe values $X_t := q_{A_t}$, and study the order statistics $X_{(1:k)} \leq \cdots \leq X_{(k:k)}$. Our goal is to compute $\mathbb{E}[X_{(j:k)}]$ exactly for any fixed $j \in \{1, \ldots, k\}$.

7.1 Reduce to the unique value support

Multiple arms may share the same value. Let the distinct values among $\{q_a\}_{a=1}^m$ be

$$v_1 < v_2 < \cdots < v_M,$$

and define the total probability mass at each value by

$$\pi_r := \sum_{a:\, q_a = v_r} p_a, \qquad r = 1, \ldots, M.$$

Then $\pi_r \geq 0$ and $\sum_{r=1}^{M} \pi_r = 1$. Define the discrete CDF at the support points

$$F_r := \mathbb{P}(X \leq v_r) = \sum_{t=1}^{r} \pi_t, \qquad F_0 := 0.$$

7.2 Distribution of the j-th order statistic

Fix $r \in \{0, 1, \ldots, M\}$. Consider the count

$$Y_r := \#\{t \in \{1, \ldots, k\} : X_t \leq v_r\}.$$

Since the X_t are i.i.d. and $\mathbb{P}(X_t \leq v_r) = F_r$, we have

$$Y_r \sim \text{Binomial}(k, F_r).$$

The event $\{X_{(j:k)} \leq v_r\}$ is equivalent to having at least j samples at or below v_r, i.e. $Y_r \geq j$. Therefore the CDF of the j-th order statistic on the discrete support is

$$\mathbb{P}(X_{(j:k)} \leq v_r) = \mathbb{P}(Y_r \geq j) = 1 - \sum_{t=0}^{j-1} \binom{k}{t} F_r^t (1 - F_r)^{k-t}. \tag{14}$$

Since $X_{(j:k)}$ only takes values in $\{v_1, \ldots, v_M\}$, its probability mass function is obtained by differencing:

$$\mathbb{P}(X_{(j:k)} = v_r) = \mathbb{P}(X_{(j:k)} \leq v_r) - \mathbb{P}(X_{(j:k)} \leq v_{r-1}) = \mathbb{P}(Y_r \geq j) - \mathbb{P}(Y_{r-1} \geq j). \tag{15}$$

7.3 Exact expectation

Combining the pmf (15) with linearity of expectation gives

$$\boxed{\mathbb{E}[X_{(j:k)}] = \sum_{r=1}^{M} v_r \Big(\mathbb{P}(Y_r \geq j) - \mathbb{P}(Y_{r-1} \geq j)\Big), \qquad Y_r \sim \mathrm{Binomial}(k, F_r).} \tag{16}$$

Special cases. For the maximum ($j = k$), $\mathbb{P}(Y_r \geq k) = \mathbb{P}(\text{all } k \text{ samples} \leq v_r) = F_r^k$, so

$$\mathbb{P}(X_{(k:k)} = v_r) = F_r^k - F_{r-1}^k, \qquad \mathbb{E}[X_{(k:k)}] = \sum_{r=1}^{M} v_r\,(F_r^k - F_{r-1}^k).$$

For the minimum ($j = 1$), $\mathbb{P}(Y_r \geq 1) = 1 - (1 - F_r)^k$, yielding

$$\mathbb{P}(X_{(1:k)} = v_r) = \big(1 - (1 - F_r)^k\big) - \big(1 - (1 - F_{r-1})^k\big) = (1 - F_{r-1})^k - (1 - F_r)^k.$$

7.4 Computation notes

- The exact computation proceeds by: (i) sorting distinct values v_r, (ii) aggregating probability masses π_r, (iii) forming cumulative sums F_r, (iv) evaluating binomial upper tails $\mathbb{P}(\mathrm{Bin}(k, F_r) \geq j)$ for $r = 0, \ldots, M$, and (v) applying (16).
- Complexity is $O(M)$ per fixed j if binomial tails are evaluated in $O(1)$ each (e.g. using a standard special function implementation), or $O(Mk)$ if computed naively via summation.
- Computing *all* $j = 1, \ldots, k$ can be done by forming the matrix of tails $T_{r,j} := \mathbb{P}(\mathrm{Bin}(k, F_r) \geq j)$ and differencing in r:

$$\mathbb{P}(X_{(j:k)} = v_r) = T_{r,j} - T_{r-1,j}, \qquad \mathbb{E}[X_{(j:k)}] = \sum_{r=1}^{M} v_r\,(T_{r,j} - T_{r-1,j}).$$

8 Conditional order statistics given that a specific arm is sampled

This section continues the *sampling with replacement* setting of Section 7. There are m arms with deterministic values $q_a \in \mathbb{R}$ and known sampling probabilities $p \in \Delta^{m-1}$. We draw k arms i.i.d. with replacement:

$$A_1, \ldots, A_k \overset{i.i.d.}{\sim} p, \qquad X_t := q_{A_t},$$

and write the order statistics of the k values as $X_{(1:k)} \leq \cdots \leq X_{(k:k)}$.

We derive the distribution and expectation of $X_{(j:k)}$ *conditioned* on the event that a particular arm i is sampled at least once.

8.1 Conditioning event and an exclusion distribution

Fix an arm $i \in \{1, \ldots, m\}$ and define the event

$$\mathcal{A}_i := \{\exists t \in \{1, \ldots, k\} : A_t = i\},$$

i.e. arm i appears at least once among the k draws. Its complement is

$$\mathcal{B}_i := \{A_t \neq i \text{ for all } t\}, \qquad \mathbb{P}(\mathcal{B}_i) = (1 - p_i)^k, \qquad \mathbb{P}(\mathcal{A}_i) = 1 - (1 - p_i)^k.$$

Conditional on $\mathcal{B}_i$, the draws remain i.i.d. but from the *renormalized distribution that excludes arm i*:

$$p_a^{(-i)} := \mathbb{P}(A = a \mid A \neq i) = \begin{cases} \dfrac{p_a}{1-p_i}, & a \neq i, \\ 0, & a = i. \end{cases}$$

8.2 A general identity for conditional events

Let E be any event measurable with respect to the k draws (equivalently, a function of $A_1, \dots, A_k$, or of $X_1, \dots, X_k$). Because $\mathcal{A}_i$ and $\mathcal{B}_i$ are complementary,

$$\mathbb{P}(E \mid \mathcal{A}_i) = \frac{\mathbb{P}(E) - \mathbb{P}(E \cap \mathcal{B}_i)}{\mathbb{P}(\mathcal{A}_i)}.$$

Moreover, since $\mathbb{P}(\mathcal{B}_i) = (1-p_i)^k$ and conditioned on $\mathcal{B}_i$ the draws are i.i.d. from $p^{(-i)}$, we have

$$\mathbb{P}(E \cap \mathcal{B}_i) = \mathbb{P}(\mathcal{B}_i)\, \mathbb{P}_{p^{(-i)}}(E) = (1-p_i)^k\, \mathbb{P}_{p^{(-i)}}(E).$$

Therefore,

$$\boxed{\mathbb{P}(E \mid \mathcal{A}_i) = \frac{\mathbb{P}_p(E) - (1-p_i)^k\, \mathbb{P}_{p^{(-i)}}(E)}{1-(1-p_i)^k}.} \tag{17}$$

An analogous identity holds for expectations of any statistic g of the k draws:

$$\boxed{\mathbb{E}[g \mid \mathcal{A}_i] = \frac{\mathbb{E}_p[g] - (1-p_i)^k\, \mathbb{E}_{p^{(-i)}}[g]}{1-(1-p_i)^k}.} \tag{18}$$

8.3 Conditional CDF, pmf, and expectation for the j-th order statistic

Apply (17) to the event $E_r := \{X_{(j:k)} \le v_r\}$, where $v_1 < \cdots < v_M$ are the distinct values taken by q_a (as in Section 7). Then

$$\boxed{\mathbb{P}(X_{(j:k)} \le v_r \mid \mathcal{A}_i) = \frac{\mathbb{P}_p(X_{(j:k)} \le v_r) - (1-p_i)^k\, \mathbb{P}_{p^{(-i)}}(X_{(j:k)} \le v_r)}{1-(1-p_i)^k}.} \tag{19}$$

Since $X_{(j:k)}$ takes values in $\{v_1, \dots, v_M\}$, the conditional mass function follows by differencing:

$$\boxed{\mathbb{P}(X_{(j:k)} = v_r \mid \mathcal{A}_i) = \mathbb{P}(X_{(j:k)} \le v_r \mid \mathcal{A}_i) - \mathbb{P}(X_{(j:k)} \le v_{r-1} \mid \mathcal{A}_i).} \tag{20}$$

Finally, applying (18) to $g = X_{(j:k)}$ yields a compact formula for the conditional expectation:

$$\boxed{\mathbb{E}[X_{(j:k)} \mid \mathcal{A}_i] = \frac{\mathbb{E}_p[X_{(j:k)}] - (1-p_i)^k\, \mathbb{E}_{p^{(-i)}}[X_{(j:k)}]}{1-(1-p_i)^k}.} \tag{21}$$

Both $\mathbb{E}_p[X_{(j:k)}]$ and $\mathbb{E}_{p^{(-i)}}[X_{(j:k)}]$ can be computed exactly using the discrete binomial-tail recipe of Section 7 after aggregating probabilities onto the distinct value support:

$$\pi_r = \sum_{a:\, q_a = v_r} p_a, \qquad F_r = \sum_{t=1}^{r} \pi_t, \qquad \mathbb{P}_p(X_{(j:k)} \le v_r) = \mathbb{P}(\mathrm{Bin}(k, F_r) \ge j),$$

and similarly for $p^{(-i)}$ (equivalently, decrease the mass of the value q_i by p_i and renormalize by $1-p_i$).

Remarks.

- The conditioning event $\mathcal{A}_i$ depends only on whether arm i is present among the k draws, not on the order. Since order statistics are symmetric functions of the multiset of values, the identities above apply directly.
- The factor $(1-p_i)^k$ is the probability that arm i is never drawn. As p_i increases or k grows, $\mathbb{P}(\mathcal{A}_i)$ approaches 1 and the conditional and unconditional quantities become similar.

9 Vectorized recipes (known p, q; sampling with replacement)

This section provides matrix-style, vectorized computations for the *exact* order-statistic expectations under i.i.d. sampling *with replacement* when p and q are known. The recipes mirror the weight-matrix constructions used earlier for sampling without replacement.

9.1 Collapse arms onto the distinct value support

Let the distinct values among $\{q_a\}_{a=1}^m$ be

$$v_1 < v_2 < \cdots < v_M,$$

and define the probability mass on each value

$$\pi_r := \sum_{a:\, q_a = v_r} p_a, \qquad r = 1, \ldots, M, \qquad \sum_{r=1}^{M} \pi_r = 1.$$

Define the discrete CDF vector $F \in \mathbb{R}^{M+1}$ by

$$F_0 := 0, \qquad F_r := \sum_{t=1}^{r} \pi_t \quad (r = 1, \ldots, M).$$

Collect values into a column vector $v := (v_1, \ldots, v_M)^\top \in \mathbb{R}^M$.

9.2 Unconditional expectations for all $j = 1, \ldots, k$

For each threshold level F_r, define the binomial *upper-tail* matrix $T \in \mathbb{R}^{(M+1)\times k}$ by

$$T_{r,j} := \mathbb{P}(\mathrm{Bin}(k, F_r) \geq j), \qquad r = 0, \ldots, M, \;\; j = 1, \ldots, k. \tag{22}$$

The pmf of the j-th order statistic on the value support is obtained by differencing the CDF in r:

$$W^{\mathrm{wr}}_{r,j} := \mathbb{P}(X_{(j:k)} = v_r) = T_{r,j} - T_{r-1,j}, \qquad r = 1, \ldots, M, \;\; j = 1, \ldots, k, \tag{23}$$

so $W^{\mathrm{wr}} \in \mathbb{R}^{M\times k}$ is a *weight matrix.*

Stacking expectations over all j yields a single matrix–vector multiply:

$$\boxed{E^{\mathrm{wr}} := \big(\mathbb{E}[X_{(1:k)}], \ldots, \mathbb{E}[X_{(k:k)}]\big) = v^\top W^{\mathrm{wr}} = v^\top \big(T_{1:M,:} - T_{0:M-1,:}\big).} \tag{24}$$

How to compute T without special functions (fully vectorized). Let $s \in \{0, 1, \dots, k\}$ index binomial counts and define the pmf matrix $P \in \mathbb{R}^{(M+1)\times(k+1)}$:

$$P_{r,s} := \binom{k}{s} F_r^s (1 - F_r)^{k-s}.$$

Then the tail matrix T is obtained by a reverse cumulative sum over s:

$$T_{r,j} = \sum_{s=j}^{k} P_{r,s}, \qquad j = 1, \dots, k.$$

Equivalently, if rcumsum denotes reverse cumulative sum along the s axis,

$$T_{:,j} = \mathrm{rcumsum}(P)_{:,j}, \qquad j = 1, \dots, k.$$

This computes *all* $T_{r,j}$ in $O(Mk)$ time using broadcasting and a single cumulative sum.

9.3 Conditional expectations given that a specific arm is sampled

Fix an arm i with probability p_i and value q_i. Let $g(i) \in \{1, \dots, M\}$ be the *value-group index* such that $q_i = v_{g(i)}$. Let $\mathcal{A}_i$ be the event that arm i appears at least once in the k draws. From Section 8,

$$\boxed{\mathbb{E}[X_{(j:k)} \mid \mathcal{A}_i] = \frac{\mathbb{E}_p[X_{(j:k)}] - (1 - p_i)^k \, \mathbb{E}_{p^{(-i)}}[X_{(j:k)}]}{1 - (1 - p_i)^k}, \qquad j = 1, \dots, k,} \tag{25}$$

where $p^{(-i)}$ is the distribution p renormalized to exclude arm i.

Value-mass update for $p^{(-i)}$. On the value support, excluding arm i modifies the masses as:

$$\pi_{g(i)}^{(-i)} = \frac{\pi_{g(i)} - p_i}{1 - p_i}, \qquad \pi_r^{(-i)} = \frac{\pi_r}{1 - p_i} \quad (r \neq g(i)).$$

Equivalently, the CDF vector transforms as

$$F_r^{(-i)} = \frac{F_r - p_i \, \mathbf{1}\{r \geq g(i)\}}{1 - p_i}, \qquad r = 0, \dots, M. \tag{26}$$

Given $F^{(-i)}$, define $T^{(-i)}$ and $W^{\mathrm{wr}(-i)}$ by the same construction as (22)–(23), and compute

$$E^{\mathrm{wr}(-i)} = v^\top W^{\mathrm{wr}(-i)} = v^\top \left(T^{(-i)}_{1:M,:} - T^{(-i)}_{0:M-1,:} \right) \in \mathbb{R}^k.$$

Then stacking (25) across all j gives the vectorized form:

$$\boxed{E^{\mathrm{wr}|\mathcal{A}_i} = \frac{E^{\mathrm{wr}} - (1 - p_i)^k \, E^{\mathrm{wr}(-i)}}{1 - (1 - p_i)^k} \in \mathbb{R}^k.} \tag{27}$$

9.4 Computing $E^{\mathrm{wr}(-i)}$ for many arms i (batch vectorization)

If conditional expectations are needed for many arms i, it is convenient to vectorize the computation of the modified CDFs $F^{(-i)}$. Let $g \in \{1, \dots, M\}^m$ be the vector of group indices $g(i)$ and let $r = (0, 1, \dots, M)$ be the CDF index. Define the *step matrix* $H \in \{0, 1\}^{m \times (M+1)}$ by

$$H_{i,r} := \mathbf{1}\{r \geq g(i)\}.$$

Then the transformed CDFs (26) can be written compactly as

$$\boxed{F_{i,r}^{(-)} = \frac{F_r - p_i H_{i,r}}{1 - p_i}, \qquad i = 1, \dots, m, \quad r = 0, \dots, M.} \tag{28}$$

Given the matrix $F^{(-)} \in \mathbb{R}^{m \times (M+1)}$, one can compute

$$T_{i,r,j}^{(-)} := \mathbb{P}(\mathrm{Bin}(k, F_{i,r}^{(-)}) \geq j),$$

by broadcasting over (i, r) and j (or by computing binomial pmfs and reverse cumsums along the count axis as described above). Differencing in r yields weights $W_{i,r,j}^{\mathrm{wr}(-)} = T_{i,r,j}^{(-)} - T_{i,r-1,j}^{(-)}$ and thus

$$E_{i,j}^{\mathrm{wr}(-)} = \sum_{r=1}^{M} v_r \, W_{i,r,j}^{\mathrm{wr}(-)}.$$

Finally, apply (27) for each i.

Practical note (memory). The fully vectorized tensor $T^{(-)} \in \mathbb{R}^{m \times (M+1) \times k}$ can be large. In practice, one typically processes arms in blocks (vectorized within a block) while reusing the same precomputed $\binom{k}{s}$ coefficients and the base CDF vector F.